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Aryabhatta Knowledge University

B.Tech 5th Semester Exam., 2020 (New Course)

Subject: COMPUTER NETWORKS & SECURITY

Code: 104501 | ***Time: 03 Hours*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option of the following (any seven): **[14]**

(a) Which transmission media provides the highest transmission speed in a network?
(i) Coaxial cable
(ii) Twisted pair cable
(iii) Optical fiber
(iv) Electrical cable

(b) The data link layer takes the packets from network layer and encapsulates them into frames for transmission.
(i) network layer
(ii) physical layer
(iii) transport layer
(iv) application layer

(c) A 4-byte IP address consists of
(i) only network address
(ii) only host address
(iii) network address and host address
(iv) network address and MAC address

(d) Network layer firewall works as a
(i) frame filter
(ii) packet filter
(iii) content filter
(iv) virus filter

(e) Which of the following is false with respect to TCP?
(i) Connection-oriented
(ii) Process-to-process
(iii) Transport layer protocol
(iv) Unreliable

(f) The header length of an IPv6 datagram is
(i) 10 bytes
(ii) 25 bytes
(iii) 30 bytes
(iv) 40 bytes

(g) Which of the following are transport layer protocols used in networking?
(i) TCP and FTP
(ii) UDP and HTTP
(iii) TCP and UDP
(iv) HTTP and FTP

(h) Which layer of the TCP/IP stack corresponds to the OSI model transport layer?
(i) Host-to-host
(ii) Application
(iii) Internet
(iv) Network access

(i) Public key encryption/decryption is not preferred because
(i) it is slow
(ii) it is hardware/software intensive
(iii) it has a high computational load
(iv) All of the above

(j) Which one of the following is not a public key distribution means?
(i) Public key certificates
(ii) Hashing certificates
(iii) Publicly available directories
(iv) Public-key authorities

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Q.2 Solve this question : **[14]**

Define computer network. Discuss various types of network topologies in computer network. Also discuss various advantages and disadvantages of each topology.

Q.3 Solve this question : **[14]**

Explain different types of errors in data transmission. Write about redundancy, detection versus correction, forward error correction versus re-transmission and coding.

Q.4 Solve this question : **[14]**

Discuss the terms 'sub-netting' and 'super-netting'. You are given the IP address of 193.103.20.0/24 and need 50 subnets. How many hosts per network and total networks do you get once subnetted?

Q.5 Solve this question : **[14]**

Explain RIP in your own words. Take a network graph (group of routers) and explain how routing is performed.

Q.6 Solve this question : **[14]**

Differentiate between TCP and UDP. Suppose a process in host C has a UDP socket with port number 6789. Suppose both host A and host B, each sends a UDP segment to host C with destination port number 6789. Will both of these segments be directed to the same socket at host C? If so, how will the process at host C know that these two segments originated from two different hosts?

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Q.7 Write short notes on the following: **[14]**

(a) SNMP
(b) FTP
(c) SMTP
(d) WWW

Q.8 Solve this question : **[14]**

Design a security service that provides data integrity, data confidentiality and non-repudiation using public-key cryptography in a two-party communication system over an insecure channel. Give a rationale that data integrity, confidentiality and non-repudiation are achieved by your solution. (Recommendation : Consider the corresponding threats in your argumentation)

Q.9 Solve this question : **[14]**

Explain public-key cryptosystem. In a public-key system using RSA, you intercept the ciphertext C=10 sent to a user whose public key is e=5, n=35. What is the plaintext M?

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